Engineering problem

The question for this project is: Can machine-learning segmentation models use RGB or RGB-D observations to predict the supplied 2D apple-tree skeleton masks, and do they provide an advantage over a simple spatial baseline? In the current world, modern agriculture and world ecology systems are using Artificial Intelligence (AI) to automate tasks such as monitoring the growth and health of plants. The main focus of trees in this report are apple trees. For an agriculture robot to be able to interact with any apple trees, information about the tree can be useful for examining the tree's growth and health or determining if the apples are ripe enough to be harvested. RGB images are able to provide information about each individual tree, while depth observations provide information about relative distance between each tree within an area. However, identifying the main structures of each tree may be difficult. During the summer season, leaves along the small branches obscure the views of the main branches, which interferes with identification of the trees. During the winter season, there won't be any leaves to obscure the main structure, which allows easier identification of the trees. Therefore, a computational method of estimating the main structure of the tree from RGB and depth observations could assist in tree identification.

As mentioned above, the engineering problem is the prediction of 2D apple-tree skeleton masks from RGB and RGB-D observations. The project focuses on the 2D binary skeleton mask that represents the main structural branches and trunk of the tree, defined by the supplied ground truths (GT). This is then treated as a binary image-segmentation problem, where each individual pixel is classified as either tree skeleton (foreground) or non-skeleton (background). A major challenge is that the ground truth skeleton only consists of an approximate 7.5% of the image, which results in an imbalance between foreground and background pixels. Additionally, the supplied RGB images have a resolution of 640 x 480 pixels, while the supplied ground truths have a resolution of 256 x 256 pixels, requiring preprocessing and alignment procedures.

The aim of this project is to develop and evaluate machine-learning methods capable of predicting 2D apple tree skeleton masks from RGB and depth observations. The investigation includes:

- a U-Net can predict 2D - tree skeleton masks from RGB images
- Modifying the loss function improves skeleton segmentation performance
- Including depth information improves prediction performance compared with RGB only input
- Trained models outperform a single Average ground truth spatial baseline generated from the training ground truth masks

For the scope of this project, it is limited to the summer observations because the ground truth skeleton masks are only available for the summer dataset, which allows for supervised training and quantitative evaluation. This project predicts 2D tree skeleton masks instead of constructing a 3D tree. The reconstruction of a 3D tree seemed to be too challenging due to the limited sampled data given. The given ground truth represents a simplified version of the

tree's main structure, the developed system should be considered as an experimental computation method for evaluating 2D skeleton prediction rather than deployment ready systems.

Inputs and output

The inputs that were provided and required for this project are the RGB images, RGB-D images and the ground truths of the apple trees. Below is a table that provides the relevant information for this project:

Data	Original Shape	Model format	Purpose	Value/Units
RGB	480 x 640 x 3	256 x 256 x 3	Image Input	Normalized to 0 to 1, originally 8 bit with values of 0-255, unitless
Depth	480 x 640 x 3	256 x 256 x 1	Additional depth input	Normalized to 0 to 1, original 8 bit with values of 0-255, physical distance units were not documented
Ground truth	256 x 256	256 x 256	Target skeleton	Originally binary values of 0 and 255, converted to values of 0 and 1, unitless

As there were a total of 562 summer datasets, all the files were matched to their corresponding tree identifiers to ensure that each RGB image is matched with the corresponding depth and ground truth images. The dataset of summer RGB, depth and GT were used for this machine learning experiment as each input observation had a corresponding ground truth mask for training and evaluation. Though the winter datasets were available, it was excluded from this experiment due to the lack of winter ground truth masks for evaluation.

Before any training began, the RGB and depth images were resized to match the resolution of the ground truth masks. As shown in the table above, the original shape of both RGB and depth images were larger than the ground truth's images. By preprocessing from the centre, it allows a more even observation between the RGB and depth images to the ground truth masks. However, this method of preprocessing may cause alignment errors as the images are only resized to fit the 256 x 256.

The datasets were then divided into three separate subsets: 393 files for training, 84 files for validation and 85 files for testing. The training set was used to train the machine learning models. The validation set was used to assess the model performance during the development and prediction threshold. The testing set was used for final performance evaluation. This split was done in order to prevent the same tree from appearing in multiple subsets so that the final model performance would be evaluated only on the testing subset, rather than the training or validation set.

The intended output for this project is a 256 x 256 binary skeleton mask, where a value of 1 indicates the tree skeleton and a value of 0 represents the background. The methods used are intended to investigate the reconstruction ability of the tree skeleton from image observations.

Computational Solution and Alternative methods

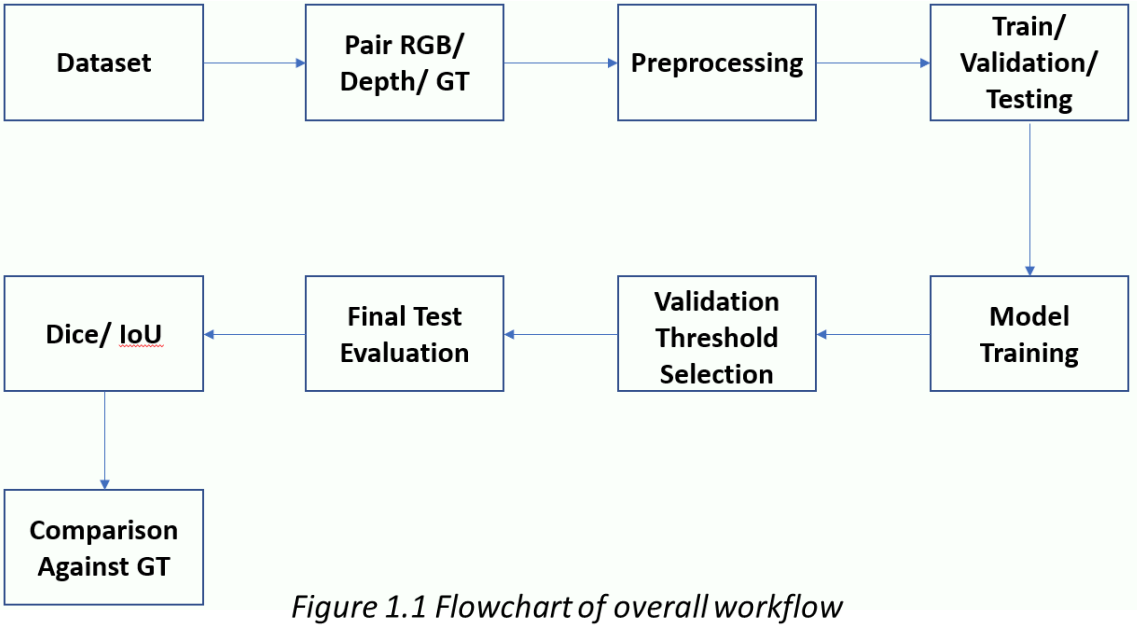


Figure 1.1 Flowchart of overall workflow

Figure 1.1 shows the overall workflow for this project. The approach taken for this project uses machine learning to predict 2D apple tree skeleton masks from RGB and depth observations. From the dataset, the RGB and depth images were paired up to their corresponding ground truth images. Then the images are preprocessed, in this case, resized to normalize and have consistent input dimensions. The prepared dataset is then divided into sub datasets of training, validation and testing. A U-net convolutional neural network is trained using the training dataset to predict the skeleton tree masks. The performance of the model was evaluated against the validation dataset, where prediction thresholds are selected based on the Dice coefficient. These models are then evaluated against the test dataset, where the performance is measured using Dice and Intersection over Union (IoU).

The reason for the use of a U-net convolutional neural network for this project is because the prediction of the tree skeleton masks is a binary image segmentation problem. The process of a U-net is it divides an image into different parts to identify specific objects. Then, it uses an encoder and decoder with the help of skip connections to extract the image features while maintaining spatial information. The encoder extracts input images, which then the spatial dimensions are progressively reduced. The decoder is then used to reconstruct the spatial resolution to generate prediction at a pixel level. Skip connections transfer information between the encoder and decoder level by level, which helps to preserve spatial details (GeeksforGeeks, 2025).

After the reconstruction of the input images by U-net, the computational model then uses binary cross entropy (BCE) loss. The model is built on U-net and BCE to develop an initial machine learning baseline to evaluate the subsequent improvements. As this project has images as an input, binary cross entropy was selected due to its ability to classify the skeleton or background as individual pixels. To help optimize the training, Adam optimizer was used as it works well with large datasets and complex models. Adam optimizer was set at a learning rate of 0.001 with a batch size of 8 epochs and a maximum of 30 epochs. To reduce unnecessary training when validation loss has stopped improving, an early stopping epoch at 5 was implemented. A prediction threshold, valued at 0.25 for this model, was selected to convert the predicted probability maps into binary masks.

As the ground truth image of the skeleton is only an approximate 7.5% of the entire image, there needs to be a way for the model to not be entirely trained on the background image. An investigation was done to determine if the segmentation performance of the original BCE loss function was able to be modified to include Dice loss. Dice loss measures the overlaps of both the RGB and ground truth, which makes it relevant to segmentation problems that involve class imbalance. Dice loss was introduced to help maintain the pixel level classification objects of BCE and is combined with BCE to help achieve greater overlap between the predicted skeleton and ground truth masks.

A second investigation was conducted to include depth observations to improve the predicted skeleton model. This model also uses the same U-net and combined BCE Dice loss function but as there is an added depth, the input channel is increased from 3 to 4. The depth channel provides information on the tree's relative depth to the observed view. This was done to determine if the addition of depth information improved the model's performance compared to pure RGB data (Neville, 2023).

A third investigation was conducted using the combination of Focal loss and Dice loss to determine whether a different approach of prediction gave a better output model. Focal loss was used as the ground truth skeleton only took up a small portion of the images, which creates a significant imbalance between the foreground and background pixels. This allows the foreground, the ground truth skeleton, to stand out more than the background, where it helps with the model prediction (Trivedi, 2020).

The last investigation was using Tversky loss, which is used due to the model over predicting the foreground after observing earlier models. This means that the previous models were producing many false positives, where a background pixel was incorrectly classified as a skeleton. For this investigation, the false positive and false negative weightings were at 0.7 and 0.3 respectively to investigate if increasing the false positive penalty weighting could reduce the foreground over prediction (Saba Hesaraki, 2023) .

The baseline average model was used to determine whether the U-net models method was more accurate than the methods than the other methods above. The equation for this baseline average model is:

$$P(x,y) = \frac{1}{N}\sum_{i=1}^{N} GT_i(x,y)$$

Where P(x,y) is the average skeleton probability at pixel (x,y), N is the number of training masks and $GT_i(x,y)$ is the binary ground truth at pixel (x,y) for training image i. The dataset that was used for this calculation was the training dataset. It was then used to create a spatial probability map that represents each pixel's frequency of the skeleton across the training dataset. Different prediction thresholds were evaluated against the validation dataset and were converted into binary masks to compare with the ground truth masks. The highest prediction threshold was used for further evaluations. This comparison is important as it allowed the project to discover if machine learning models could outperform a typical skeleton structure from the datasets.

The computational differences between the Dice model and the average GT baseline were not saved nor recorded, which limits the ability to compare the computational efficiency quantitatively. U-Net models using BCE + Dice loss required more computational effort than Average- GT baseline. As shown below, it requires an iterative training and image specific predictions for its output, thus scalability requires extra iterative training as datasets increase in size. From the equation above, the average GT baseline only requires the ground truth masks and binary ground truth and averaging the results with the number of data, thus making scalability easier. However, average GT baseline scalability is only easier under the mentioned conditions.

Validation Method

As mentioned above, the datasets were divided into training, validation and testing. This was done to minimize the same input being present in all three cases. The training dataset was used to train U-net models and the validation dataset was used to compare the model's performance during training and development to determine the optimal prediction threshold for the Dice model. The testing dataset was used to compare against the model to observe the models' ability to predict skeleton masks from unseen data.

The Dice coefficient was used to assess the overlap between the prediction and ground truth skeleton masks. The equation used here is:

$$Dice = \frac{2TP}{2TP+FP+FN}$$

Where TP is true positives, which are accurately predicted skeleton pixels, FP is false positives, which are incorrectly predicted background pixels and FN is false negatives, which are incorrectly predicted skeleton pixels as background. The output dice ranges from 0 to 1, where 0 means there is no overlap and 1 means there is a perfect overlap. This was selected as a validation method as the skeleton pixels only occupy a small part of the image, which helps provide an overview of the overlap between the predicted and skeleton mask (Lathashree Harisha, 2023).

The second validation method that was used is the Intersection over Union (IoU). The equation is similar to the Dice coefficient above:

$$IoU = \frac{TP}{TP+FP+FN}$$

IoU is used more to measure the overlap between the predicted and ground truth skeleton regions that is relative to the total area classified as a skeleton by either masks, whereas Dice is used for a specific region, which is the skeleton mask in this case. A value of 0 and 1 were given for each individual pixel of the U-Net models. It determines whether the pixels were part of the skeleton or not. A prediction threshold is applied to convert the probability of the pixels into binary skeleton masks where a 0 represents the background and 1 represents the skeleton.

17 tests were done across the python test files and all of them passed, providing evidence that the python files behaved as intended. These python files include loss functions, evaluation metrics, dataset pairing, preprocessing and dataset splitting.

Results and Comparison

Method	Threshold	Dice	IoU
RGB BCE	0.25	0.3543	0.2171
RGB BCE + Dice	0.3	0.3658	0.2260
RGB Depth BCE + DICE	0.35	0.3735	0.2319
Average-GT	0.2	0.3792	0.2384
Focal + Dice	0.45	0.3670	0.2272
Tversky	0.5	0.1406	0.0757

Table 1.1 Validation performance of each prediction methods with U-Net

Table 1.1 highlights the validation performance results of each different method used in this project. As mentioned above, the comparison tool used was the Dice coefficient and Intersection over Union (IoU). Higher values indicate that there is better overlap between the predicted and ground truth masks whereas lower value indicates worse overlap between predicted and ground truth masks. The Tversky model produced the lowest performance out of all the other methods, with a Dice coefficient of 0.1406 and an IoU of 0.0757. For the U-net models, RGB Depth BCE + Dice performed the best with a Dice coefficient of 0.3735 and an IoU of 0.2319. The highest performance was the Average-GT method, with a Dice coefficient of 0.3792 and an IoU of 0.2384. This suggests that the Tversky model performed poorly to overlap the ground truth masks with its predicted model. However, interestingly enough, Average-GT had the highest performance as it does not use any RGB or depth

information to help predict the ground truth masks, instead it uses the spatial distribution of skeleton pixels from the training masks to calculate the individual pixel's frequency.

Method	Threshold	Dice	IoU
RGB BCE	0.25	0.3506	0.2149
RGB BCE + Dice	0.3	0.372018	0.231315
RGB Depth BCE + Dice	0.35	0.371957	0.231123
Average_GT	0.20	0.3796	0.2397

Table 1.2 Testing performance of each prediction methods with U-Net

Table 1.2 represents the testing performance of the selected methods. The prediction thresholds were kept unchanged to test the model's ability to predict on unseen data. From the table, RGB BCE + Dice and RGB Depth BCE + Dice achieved similar results where the Dice coefficient is an approximate 0.3720. Both these models outperformed the base model of RGB BCE that had a Dice coefficient of 0.3506 and IoU of 0.2149. Another surprising result is the Average-GT having a higher Dice coefficient of 0.3796 and an IoU of 0.2397. This suggests that a simple baseline training model is as effective as a U-Net model. Additionally, Focal + Dice and Tversky models were investigated with the validation set but were not trained against the test data as Focal + Dice performed lower than RGB Depth BCE + Dice. Tversky performed significantly worse than all the methods.

To determine the success rate of using Dice loss, taking the difference between RGB BCE + Dice and RGB BCE gives us:

$$\Delta Dice = 0.3720 - 0.3506 = 0.0214$$

This indicates that including Dice loss improves the skeleton prediction model. As Dice loss encourages overlapping between predicted and ground truth masks, it improves the ability of the method to model the tree skeleton. Though this improvement is barely noticeable as the difference is very small.

To determine the effect of Depth information, taking the difference between RGB BCE + Dice and RGB - Depth BCE + Dice gives us:

$$\Delta Dice = 0.371957 - 0.372018 = -0.000061$$

This indicates that the depth information had a miniscule effect on the output model. Factors that could've contributed to this are the limited dataset size, the preprocessing of input images to match the ground truth resolution and just pure images of depth data without the company of solid data.

Method	Predicted foreground relative to Ground Truth
RGB BCE + Dice	2.577
RGB - Depth BCE + Dice	2.666
Focal + Dice	2.576
Tversky	13.278

Table 1.3 Foreground prediction

Further analysis of the output models were conducted to determine the foreground overprediction regions of the skeleton masks, shown in *Table 1.3*. The given ground truth masks data has an approximate 7.5% of foreground pixels, the RGB BCE + Dice model gave a prediction of an approximate of 19.3%, which was calculated by the following example:

$$7.5\% \ * \ 2.577 \ \approx \ 19.3\%$$

Though most of the output models gave similar results with RGB - Depth BCE + Dice coming in at 2.666 and Focal + Dice with 2.576, Tversky demonstrated a highly poor performance by producing a much higher overprediction value of 13.278 relative to the ground truth. Even though Tversky was shown to overpredict the ground truth, all the methods used overpredicted actual ground truth masks.

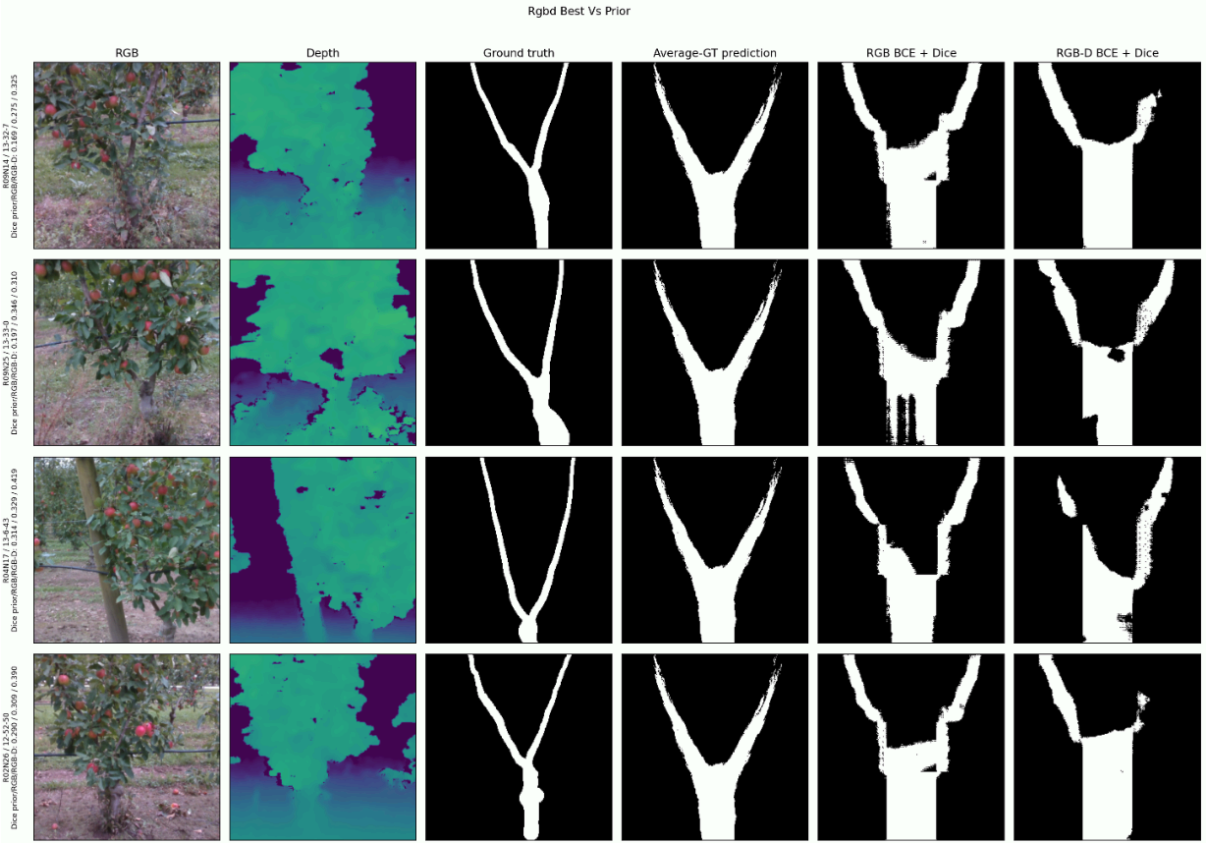


Figure 1.2 Prediction model output comparison

Figure 1.2 represents the prediction models when evaluated against unseen data. As shown on the fourth, fifth and sixth column, the predictions of these models do not represent the tree skeleton closely at all. Large blocky main trunks and branches are present as well as similar looking models too. As shown, U-net predictions seem to follow a common spatial pattern among the tree skeletons whereas the average GT baseline produced a more comparable result even though the output tree skeletons are similar. This reinforces the finding that complex models like U-net do not have an advantage over average GT baseline under conditions.

The most important finding result was the Average-GT baseline achieving the highest performance during the testing phase. As mentioned, the method for calculating the prediction was using a spatial probability map that represents each pixel's frequency of the skeleton across the training dataset, without the use of RGB and depth information inputs. As shown in *Table 1.2* the results indicate that the trained U-Net models did not perform as well as the Average-GT baseline. Yet, this does not reflect that the models failed to learn any image specific features as these models performances' are evaluated against tested conditions.

Optimisation and performance

The optimization and performance process started with the U-Net model using RGB images and BCE losses. Depth information was used to investigate whether additional spatial information could improve performance. Dice loss was used to improve the overlap between predicted and ground truth skeleton masks. Methods to compare the output models are the foreground overprediction, alternative loss functions, including Focal + Dice and Tversky. These models were then compared to the Average-GT baseline to evaluate their performances.

As shown in *Table 1.1*, The BCE model had a Dice coefficient of 0.3506 and with the added Dice loss to the original BCE model, it improved to 0.3720. This suggests that having overlaps between the prediction and skeleton masks were beneficial to the training. Focal + Dice and Tversky were also evaluated against the validation data, however, Tversky performed poorly due to overprediction of foreground while Focal + Dice was not used due to excess data to choose from.

Then the depth information was used to determine whether incorporating it into the models affected the outputs. As mentioned above, it barely affected the performance of the U-Net models.

The Average-GT baseline was used to determine if the use of U-Net models for this project was justified due to their complexity. Even though this model did not use the inputs of RGB and depth information, it had a higher Dice coefficient compared to the RGB models. This suggests that patterns within the ground truth data were sufficient enough to achieve similar

or better results, which highlights the importance of simpler alternative computational methods.

AI use and reflection

The AI tools used for this project were ChatGPT and Github Copilot and were used to help support the machine learning development, programming and documentation. ChatGPT was used to help understand the concepts of the U-Net method, loss functions, evaluation metrics and dataset validation. GitHub Copilot was used to help with writing the code, debugging and documentation for GitHub

The decisions from AI were tested through running the python scripts and checking the outputs of these scripts. A total of 17 tests were done and all of them were successful. These tests cover preprocessing, dataset pairing, loss functions, evaluation metrics and dataset splitting. The models were then assessed using the Dice and IoU methods.

The responsibilities that were under the student's task were defining the project's scope, the raw data inputs, selecting the objective of the 2D prediction and interpreting the experiment results.

A limitation of the AI in this project is that it sometimes goes off track with the things I want to do. Originally the project idea was modelling a 2D tree using the help of the images, however, I did not notice the AI was going off track until running the python scripts. Then I had to change my project scope from modelling a 2D tree to the ability of machine learning models using RGB and depth information to predict 2D apple tree skeleton masks.

Through this project, I've learnt that I need to be very precise and continuously remind AI what we are doing and what we are trying to achieve in this project. I've learnt how AI is able to provide image learning models that allowed me to do this investigation. This allowed me to use the Dice coefficient to answer the proposed question at the start. AI was also helpful to understand these methods and how certain methods may be better in certain situations than complicated models, in this case Average-GT baseline was better than Dice coefficient due to the set conditions. Finally, AI was useful to help generate code and explain unfamiliar concepts, however, everything from AI must be evaluated first before running to ensure that the correct codes are running.

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Appendices

Project GitHub Link:
https://github.com/javieryiiii/MMA3001Project.git